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Research article

Thermodynamics, kinetics and thermal decomposition characteristics of sewage sludge during slow pyrolysis

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ABSTRACT

Pyrolysis has shown great potential for sewage sludge valorisation and management by producing value-added chemicals. Although the product process yields are extensively studied, a few studies exist without consensus on the kinetic properties of sewage sludge pyrolysis. As a result, a study to investigate the thermal decomposition characteristics of Gauteng sewage sludge (GSS) at various heating rates (10, 20, and 30 °C/min), its pyrolysis kinetic parameters, reaction mechanism and thermodynamic properties was meticulously conducted. The results show that sewage sludge decomposition occurs in three stages, whereby the main decomposition (active pyrolysis) takes place at 150–570 °C. Fourier transform infrared spectroscopy (FTIR) analysis results confirm progression of thermal decomposition of GSS and drive off volatile compounds and formation of aromatic structures during TGA studies of GSS. An increase in heating rate shifts the characteristic temperatures towards higher temperatures with the highest decomposition rate of 1.10%/min.mg at 30 °C/min. The activation energies of GSS pyrolysis were calculated using Flynn-Wall-Ozawa, Kissinger-Akahira-Sunose and Starink methods and averaged as 225.92, 218.04 and 218.97 kJ/mol, respectively. GSS pyrolysis involves complex reaction chemistry with high reactivity whereby reactions that follow third order and three-dimensional diffusion-reaction mechanisms dominated the process. However, these mechanisms cannot be used explicitly to define the global pyrolysis kinetics due to the occurrence of multiple simultaneous reactions. The obtained thermodynamic and kinetic data will advance and amplify the design, simulation and optimisation of global energy pyrolysis units for production of value-added chemicals.

1. Introduction

GSS which is produced in South Africa is an inevitable by-product of biological wastewater treatment plants (WWTPs). The latest South African sewage sludge tonnage report indicates at least 700ktons (dry solids) were generated from only 55% urban population in 2011 (Department of Environmental Affairs, 2012). As a developing country, an increase in the amount of GSS is envisaged with increased urbanisation and population in the near future. Presently, GSS is disposed of via landfill, agricultural activities and incineration (Mphahlele, 2020). However, strict legislation (National Environmental Management: Waste Act No. 59 of 2008) restricts disposal of GSS by conventional methods (Government Gazette, 2008). Also, incineration of sewage sludge causes atmospheric emissions of volatilised metallic elements (Gerstle et al., 1982). As a result, there is a need for the development of technology that utilises GSS in a manner that is an environmentally friendly and economically viable.

Several thermochemical processes such as incineration, combustion, liquefaction, gasification and pyrolysis have gained attention over years as potential technologies to economically manage and valorise sewage sludge (Fonts et al., 2009). However, incineration and combustion are restricted by their high proportions of heavy metallic element volatilisation, while liquefaction operates at low temperatures which may not be sufficient to provide the required energy to break complex organic structures of sewage sludge (Udayanga et al., 2009). South Africa has available commercial coal gasifiers which might be useful for GSS conversion into valuable chemicals with a reduced capital cost.

Pyrolysis has gained global interest over the years as a potential technology for disposal and extraction of energy from sewage sludge due to its ease of product selectivity and the ability to produce useful pyrolytic products such as the aqueous pyrolysis condensate, condensed

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Nomenclature

A	Pre-exponential factor (s^{-1})
E_a	Activation energy (kJ/mol)
ΔG	Gibbs free energy (kJ/mol)
ΔH	Enthalpy (kJ/mol)
h	Planks constant (m^2 kg s^{-1})
K_B	Boltzmann constant (m^2 kg $s^{-2}K^{-1}$)
R	Universal gas constant (J $K^{-1}mol^{-1}$)
R^2	Coefficient of determination (−)
ΔS	Entropy (kJ/mol)
T	Temperature (K)
t	Time (s)
T_m	Peak decomposition temperature (K)
m_0	initial sample mass (mg)
m_∞	sample mass at the final temperature (mg)
m_T	sample mass at a specific temperature (mg)
x	Conversion (−)
β	Heating rate (K/min)

organic liquid, non-condensable gas and a stable char (Fonts et al., 2009; Shahbeig and Nosrati, 2020). Other researchers from around the world have made great efforts to study the production of energy, fuel and value-added chemicals through pyrolysis of sewage sludge in pilot and bench-scale units (Alvarez et al., 2015; Park et al., 2008; Udayanga et al., 2019).

Fonts et al. (2009) studied fast pyrolysis of sewage sludge from three distinct Spanish WWTPs in a fluidised-bed reactor. The composition of the sewage sludge samples differed due to the unstandardized treatment methods, consequently affecting the yields and compositions of the pyrolytic products. The same authors also reported that samples with a higher percentage of ash yields produced larger proportions of gaseous streams due to the catalytic effect of some alkali and alkaline elements present in the mineral matter. The liquid products (aqueous pyrolysis condensate + organic liquid) comprise mainly of oxygenated compounds and trace amounts of solid particles which accelerate ageing of the product.

Gomez-Rico et al. (2005) investigated the combustion and pyrolysis behaviors of 17 Spanish sewage sludge samples using TGA and digital scanning calorimeter (DSC). The authors report parallel thermal profiles under pyrolysis and combustion conditions until 600 °C suggesting combustion to be an oxidative pyrolysis. The authors further indicate that combustion of sewage sludge is mainly an exothermic process.

Alvarez et al. (2015) performed fast pyrolysis of sewage sludge from Barcelona (Spain) in a conical spouted reactor to reduce the yield of oxygenated and entrained solid particles in the liquid product by reducing the solid residence time. The reduced heat and mass transfer limitations of the conical spouted reactor subsequently reduced the content of oxygenates and the number of entrained solids in the pyrolytic liquids. Also, the formation of ammonia, methanethiol, carbon dioxide and hydrogen were slightly reduced by an effective char removal dynamic of the reactor.

Mphahlele (2020) and Park et al. (2008) studied GSS and Korean sewage sludge, respectively. Both samples were anaerobically digested, and a polymer flocculant was used during the dewatering process. However, the composition of sewage sludges differed significantly indicating that the source of wastewater also has an impact on the composition of sewage sludge even when treatment methods are similar.

Udayanga et al. (2009) studied the effect of mineral matter and organic matter of Singaporean sewage sludge on the reaction mechanism, product yields and quality. The authors reported carbon dioxide sequestration and ring-opening of oxygenated compounds by calcium oxide during pyrolysis. Immobilisation of heavy metallic elements by humic acid and calcium oxide was also observed. From this perspective, it is evident that the source of sewage sludge affects its compositions and its pyrolysis reaction pathways, which highlights the need to understand the reaction mechanism of GSS and assess its suitability as a pyrolysis/gasification feedstock.

At this point, industrial application of sewage sludge pyrolysis is still far from its final stage of implementation and for that to happen enough literature should be developed. Also, it is essential to understand the kinetic and thermodynamic properties of GSS pyrolysis for plant design and optimisation. However, there is limited literature on the kinetic parameters and no reports on thermodynamic properties of GSS pyrolysis, which the current study aims to address.

Rodriguez et al. (2008) performed pseudo-mechanistic kinetic studies of sewage sludge sampled from the San Juan wastewater treatment plant in Argentine city under inert and oxidising conditions. Although this technique correlated the results well, a 2% over estimation of kinetic parameters is observed at temperatures around 660 °C and slower heating rates (~5 °C/min). It is important to meticulously select the operating conditions such as reactor configuration, heating and particle size for effective pyrolysis of biomass and for the estimation of kinetic parameters (Van de Velden et al., 2008). Biomass composition varies based on geographical locations. As a result, data from literature needs to be used with caution to avoid overestimation of kinetic and thermodynamic parameters in the design space (Van de Velden et al., 2008).

Kinetic parameters are important for designing and sizing of treatment units and also for an environmental and economic feasibility of pyrolysis technology for the sewage sludge management. Iso-conversional methods such as Flynn-Wall-Ozawa (FWO), Kissinger-Akahira-Sunose (KAS) and Starink have been used widely to define the kinetic properties of pyrolysis reactions (Fernandez-Lopez et al., 2016; Maurya et al., 2016; Slopiecka et al., 2012) potentially due to their high coefficient of determination (R^2) for a variety of pyrolysis feedstocks. Also, their ability to predict activation energy as a function of conversion without making model assumptions has made these techniques favourable (Yuan et al., 2017). However, it is also important to understand the reaction mechanism of which the solid-state reactions followed during pyrolysis. One way of predicting the reaction mechanism is by using the Criado (1978) z-master plots method. This technique involves the comparison of the integral and derivative functions of conversion as defined by the Coats-Redfern procedures (Coats and Redfern, 1964).

The main objective of this study was to determine the thermal characteristics, thermodynamic properties, kinetic parameters, and reaction mechanisms of the GSS pyrolysis using three iso-conversional model-free methods (FWO, KAS and Starink), model based models (Coats-Redfern method) and Criado (1978) z-master plots, which was not investigated in the past. There are limited literature on thermogravimetry (TGA) and differential thermogravimetry (DTG) profiles, thermodynamic properties, kinetic parameters, and reactions of the GSS pyrolysis. TGA was used for studying the pyrolysis characteristics because of its high sensitivity to mass and temperature and the ability to provide step-wise decomposition data using minimum sample (5–20 mg) (Singh et al., 2012). The FTIR analysis of the GSS samples was employed to better understand the organic structural changes and volatilisation of semi and volatile organic compounds occurred during pyrolysis. TGA gives mass loss information as a function of temperature, and this information can be used to determine the kinetic parameters and thermodynamic properties of the GSS pyrolysis. The results obtained can be used for the design and optimisation of global energy pyrolysis plants using simulation software techniques.

2. Materials and methods

2.1. GSS sampling and preparation

GSS subjected to anaerobic digestion and dewatering was sampled from a domestic WWTP (Fig. 1) in Gauteng Province of South Africa and used in this study. A 80 kg sample was taken on a daily basis for one month to produce a composite of 2400 kg for the sample preparation following international organisation for standardisation (ISO) standards, ISO 1988:1975 and ISO13909–4:2016 sampling procedures. The total representative sample of 80 kg was dried in 2 parts in an oven (Scientific Incubator: 240) at 105 °C for 24 h under the inert atmosphere (N_2). About 40 kg of sample was obtained after oven drying due to moisture loss.

The dried samples (40 kg) were ground to obtain <75 μm particles and split into small representative samples of 1 kg using a rotary sample divider. Particles of <200 μm have been reported to have lumps of uniform temperatures (Van de Velden et al., 2008), hence <75 μm particle size was selected to minimise the effect of heat transfer resistance within an individual particle. The prepared homogenous samples were stored under an inert atmosphere of 99.995% nitrogen gas (N_2) which was supplied by African Oxygen (Afrox) Ltd, South Africa for the analyses and tests. 99.995% N_2 was also used as a carrier gas and an inert atmosphere medium during the thermogravimetric analysis.

2.2. GSS characterisation

Proximate and ultimate analyses of the representative GSS sample were conducted with respect to the coal standard procedures ISO 17246:2010 and ISO 17247:2020 respectively, and reported elsewhere by Mphahlele (2020) and Tsemane et al. (2019). The mineralogical and chemical properties of GSS have been performed and described elsewhere by Mphahlele (2020) and Tsemane et al. (2019), following the analytical standard methods which are described by Norrish and Hutton (1969), Rietveld (1969) and Speakman (2012). FTIR analysis of GSS and chars obtained at different temperatures was performed using PerkinElmer spectrum 2 IR and PerkinElmer spectrum software (v.10.5.4.738). The IR analysis was performed with a wavenumber range from 4500 to 500 cm^{-1}, a resolution of 4 cm^{-1} and 25 scans.

2.3. TGA analysis

TGA analysis of the representative GSS sample was carried out using a PerkinElmer Pyris 1 TGA: N5370031 at three heating rates (10, 20 and 30 °C/min) under an inert environment of N_2. N_2 flowing at 100 ml/min was used to maintain an oxygen-deficient environment and as a sweep gas. Blank experiments were carried out prior to sample experiments for noise cancellation (baseline correction). For each experiment, 10 mg ± 1 mg of GSS was placed in a platinum pan and heated from ambient temperature to 850 °C. All experiments were repeated three times at each heating rate, and the averages were taken. The rate of mass loss obtained from the TGA data was used to calculate the DTG and determine the kinetic parameters by applying KAS, FWO and Starink methods along with studying the reaction mechanisms using Microsoft Excel 365 software.

2.4. Kinetic triplet

Thermal decomposition of carbonaceous materials involves a series of reactions occurring simultaneously, which makes it challenging to describe the reaction mechanism by assessing individual reactions. An easier approach involves the use of overall mass loss which can be determined using Equation (1), which defines the degree of conversion (x) of the sewage sludge to pyrolysis products.

$$x = \frac{m_0 - m_T}{m_0 - m_\infty} \tag{1}$$

where m_0 is the initial sample mass, m_T is the sample mass at a specific temperature (T) and m_∞ is the sample mass at the final temperature. Since the thermogravimetric study is carried out under non-isothermal conditions with a constant heating rate $\beta = dT/dt$, the rate of decomposition is defined by two functions: the temperature function $k(T)$ and the conversion function $f(x)$. Equation (2) shows the relationship thereof.

$$\frac{dx}{dt} = f(x)\,k(T) \tag{2}$$

The Arrhenius equation shows that the temperature dependency of the reaction can be defined by the rate constant (Equation (3));

Fig. 1. A typical process in Gauteng for treatment of municipal wastewater (Frankson, 2018).

$$k(T) = Ae^{\left(-\frac{E_a}{RT}\right)} \tag{3}$$

where:

k(T) is the reaction rate constant (s^{-1}); A is the frequency factor (s^{-1}); Ea is the activation energy (J/mol);

R is the universal gas constant (J/mol.K); and.

T is the temperature (K).

Incorporating the heating rate (β) and substituting Equation (3) into Equation (2), Equation (4) is obtained.

$$\frac{dx}{dT} = \frac{A}{\beta}\exp\left[-\frac{E_a}{RT}\right]f(x) \tag{4}$$

After integrating Equation (4), Equation (5) is obtained.

$$g(x) = \int_0^x \frac{dx}{f(x)} = \frac{A}{\beta}\int_{T_0}^{T}\exp\left[-\frac{E_a}{RT}\right]dT = \frac{AE_a}{\beta R}p(u) \tag{5}$$

whereby, $p(u) = \int_0^\infty \exp(-u)/u^2\ u = E_a/RT$. The limits of the temperature integral are based on the assumption that no reaction occurs at $T = 0$ and the point where pyrolysis starts ($T = T_0$).

2.5. Iso-conversional methods: KAS, FWO and starink

Since there is no explicit solution for $p(u)$, KAS, FWO and Starink methods were developed based on different techniques for approximations of $p(u)$(Dhyani and Bhaskar, 2018). Substituting the value of $p(u)$ into Equation (5) and linearising leads to a simplified correlation (Equation (6)), which incorporates the various approximations in the form of constants (S and B).

$$\ln\left[\frac{\beta}{T_x^S}\right] = C_s - \left[\frac{BE_a}{RT_x}\right] \tag{6}$$

whereby $S = 0$ and $B = 0.457$ for FWO, $S = 2$ and $B = 1$ for KAS and $S = 1.8$ and $B = 1.0037$ for Starink. Since $\frac{BE_a}{R}$ is constant, a plot of $\ln\left|\frac{\beta}{T_x^S}\right|$ against $\frac{1}{T}$ should be linear with $-\frac{E_a}{R}$ as the slope and constant (Cs). Therefore, from the slope, E_a can be calculated. However, a high coefficient of determination (R^2) must be obtained from the linear graph for the satisfactory confidence on the calculated value of E_a.

2.6. Model fitting method: Coats-Redfern technique

Coats-Redfern equation is based on Rainville function for solving $p(u)$. The linearised form of the correlation is shown in Equation (7).

$$\ln\left[\frac{g(x)}{T^2}\right] = \ln\left[\left(\frac{AR}{\beta E_a}\right)\left(1 - \frac{2RT}{E_a}\right)\right] - \frac{E_a}{RT} \tag{7}$$

However, $2RT/Ea << 1$ in the temperature range commonly used for pyrolysis which made it sensible to assume $\ln[(AR/E_a)(1 - 2RT/E_a)]$a constant. A plot of $\ln[g(x)/T^2]$ against $1/T$ was obtained with slope $- E_a/R$. The value of the frequency factor (A) was obtained by equating the constant to $\ln[(AR/\beta E_a)]$ and solving.

2.7. Generalised z-master plots: criado technique

Criado z-master-plots method was employed for determining the reaction mechanisms taking place during the GSS pyrolysis (Dhyani and Bhaskar, 2018). The method determines the characteristics of each solid-state reaction mechanism by comparing the reduced theoretical curve ($f(x)g(x)/f(0.5)g(0.5)$) with the reduced reaction rate obtained from the experimental results ($(T_x/T_{0.5})^2 \times (dx/dt)/(dx/dt)_{0.5}$). Equation (8) shows the relationship between the two curves (master plots). The descriptions of the Integral ($f(x)$) and differential ($g(x)$) mathematical representation of the reaction progression mechanisms are illustrated in Table 1.

$$\frac{Z(x)}{Z(0.5)} = \frac{f(x)g(x)}{f(0.5)g(0.5)} = \left(\frac{T_x}{T_{0.5}}\right)^2 \times \left(\frac{dx/dt}{dx/dt_{0.5}}\right) \tag{8}$$

Conversion $x = 0.5$ is used as the reference point where all master plots of the assumed reaction models intercept at $Z(x)/Z(0.5) = 1$. The reaction model of best fit is the one with great proximity to the experimental master plot.

2.8. Thermodynamic properties

Thermodynamic properties such as Gibbs free energy (ΔG), entropy change (ΔS), enthalpy change (ΔH), and frequency factor (A) of any reaction are essential to understand the energy potential, and distribution, and these properties can be obtained by substituting the calculated values of E_a in Equations (9)–(12) (Mallick et al., 2018). Vyazovkin et al. (2011) indicated the use of a compensation method to minimise the errors when determining the pre exponential factor. However, this technique involves advanced computation with somewhat a minimal

Table 1

Integral and differential functions for solid-state single heating rate reaction models.

Models	Code	f(x)	g(x)
Order based models			
First-order	R1	$(1-x)$	x^2
Second-order	R2	$(1-x)^2$	$(1-x)\ln(1-x) + x$
Third-order	R3	$(1-x)^3$	$\frac{1}{2}[(1-x)^{-2} - 1]$
Diffusion models			
one-dimensional diffusion	D1	$\frac{1}{2x}$	x^2
Two-dimensional diffusion (Valensi model)	D2	$[-\ln(1-x)]^{-1}$	$(1-x)\ln(1-x) + x$
Three-dimensional diffusion (Jander model)	D3	$\frac{3}{2}(1-x)^{\frac{2}{3}}\left[1-(1-x)^{\frac{1}{3}}\right]^{-1}$	$[1-(1-x)^{\frac{1}{3}}]^2$
Three-dimensional diffusion (Ginstling model)	D4	$\frac{3}{2}\left[(1-x)^{\frac{1}{3}} - 1\right]^{-1}$	$1 - \frac{2x}{3} - (1-x)^{\frac{2}{3}}$
Nucleation or growth models			
Power law, n = 2/3	P2/3	$\frac{2}{3}x^{-\frac{1}{2}}$	$x^{\frac{3}{2}}$
Power law, n = 1/2	P2	$2x^{\frac{1}{2}}$	$x^{\frac{1}{2}}$
Power law, n = 1/3	P3	$3x^{\frac{2}{3}}$	$x^{\frac{1}{3}}$
Power law, n = 1/4	P4	$4x^{\frac{3}{4}}$	$x^{\frac{1}{4}}$
Avrami-Erofeev, n = 1	A1	$\frac{3}{2}(1-x)[-\ln(1-x)]^{\frac{1}{3}}$	$[-\ln(1-x)]^{\frac{2}{3}}$
Avrami-Erofeev, n = 2	A2	$2(1-x)[-\ln(1-x)]^{\frac{1}{2}}$	$[-\ln(1-x)]^{\frac{1}{2}}$
Avrami-Erofeev, n = 3	A3	$3(1-x)[-\ln(1-x)]^{\frac{2}{3}}$	$[-\ln(1-x)]^{\frac{1}{3}}$
Avrami-Erofeev, n = 4	A4	$4(1-x)[-\ln(1-x)]^{\frac{3}{4}}$	$[-\ln(1-x)]^{\frac{1}{4}}$
Random Nucleation	F4	$(1-x)^2$	$\frac{1}{(1-x)}$
Random Nucleation	F5	$\frac{(1-x)^3}{2}$	$\frac{1}{(1-x)^2}$
Sigmoidal rate equations			
Prout-Tomkins	F1	$x(1-x)$	$-\ln(1-x)$
Geometric contraction models			
Contracting cylinder	F2	$2(1-x)^{\frac{1}{2}}$	$1-(1-x)^{\frac{1}{2}}$
Contracting sphere	F3	$2(1-x)^{\frac{1}{3}}$	$1-(1-x)^{\frac{1}{3}}$

benefit. As a result, the method described in Mallick et al. (2018) has been used. However, the reader should take caution when using this data for design and simulation of the pyrolysis units.

$$A = \beta E_a \exp\left[\frac{E_a}{RT_m}\right] \Big/ RT_m^2 \tag{9}$$

$$\Delta H = E_a - RT \tag{10}$$

$$\Delta G = E_a + RT_m \ln\left(K_B T_m / hA\right) \tag{11}$$

$$\Delta S = \frac{\Delta H - \Delta G}{T_m} \tag{12}$$

whereby K_B is the Boltzmann constant, T_m is the temperature at maximum rate of breakdown and h is the Planck constant.

3. Results and discussion

3.1. FTIR analysis

The FTIR spectrum of the chars obtained at different heating rates indicates an insignificant difference (Intani et al., 2018), as a result the spectrum of only one char sample was discussed and compared to the GSS spectrum. Table 2 illustrates the FTIR bands and assignments for the GSS and char samples for the functional groups and fingerprints made using literature (Georgakopoulos et al., 2003; Morrison and Boyd, 1987; Zhang et al., 2011).

Table 2
A list of FTIR band positions and assignments for GSS and one char sample obtained at 850 °C final temperature and 30 °C/min heating rate.

Band position (cm^{-1})		Band assignments	References
GSS	Char		
3700–3600	3700–3600	O–H Stretch: Hydroxyl group by water of hydration	Georgakopoulos et al. (2003)
3550–3200	–	O–H Stretch: intermolecular –H bonds by phenols and alcohols	(Han et al., 2016)
3500–3300	–	N–H stretching vibrations: aromatic amines, primary amides and amines (free N–H)	Georgakopoulos et al. (2003)
3350–3010	–	N–H stretching vibrations: secondary amines (hydrogen bonded N–H)	Georgakopoulos et al. (2003)
3036–2942	–	C–H stretching vibrations: low-molecular aliphatic structures, probably alkanes	(Pavia et al., 1996; Meng et al., 2014)
2960–2850	–	C–H stretching vibrations: aromatic structures	(Leng et al., 2015)
1700–1720	–	C=O–O carboxylic acids	Fonts et al. (2009)
1740–1658	–	COH stretching vibrations: by aldehydes and amides	Zhang et al. (2011)
1600–1500	1600–1500	C=C and ***C*** ⋯ ***C***: Alkenes and aromatic rings	Georgakopoulos et al. (2003)
1590–1580	–	Benzene ring in aromatic hydrocarbons	Fonts et al. (2009).
1385–1345	–	–NO_2 bending vibrations: Nitro compounds	Georgakopoulos et al. (2003)
1200–900	1200–900	Si–O–Si vibrations: quartz and silica based minerals	(Fonts et al., 2009; Pavia et al., 1996).
1150–1000	–	Polysaccharides, carboxylic acids	(Fonts et al., 2009; Georgakopoulos et al., 2003)
1060–990	–	Secondary cyclic alcohols	Georgakopoulos et al. (2003)
990–450	990–450	C-x vibrating streches: carbon-halides/organically assoiciated inorganics	Fonts et al. (2009)
–	850–700	C–H vibrations: Aromatic rings	(Leng et al., 2015; Zhang et al., 2011)

-: no visible peak.

In the case of GSS, the two peaks at 3700-3600 cm^{-1} and 3550-3200 cm^{-1} were caused by the OH stretching vibrations, possibly due to water of hydration, and phenolic compounds, respectively. The functional group assignments agree with solvent extraction experiments followed by GC-MS scan analysis of GSS which indicated the presence of aliphatic (3036-2942 cm^{-1}), aromatic (2960–2850, 1600-1500 cm^{-1}), phenolic (3550-3200 cm^{-1}) and alcohol (3550–3200 and 1060-990 cm^{-1}) compounds in this sample (Kotowska et al., 2012; Mphahlele, 2020). A peak at 1700-1720 cm^{-1} is attributed to carboxyl functional group, possibly of polysaccharides (1150-1000 $cm^{-1)}$ originating from faecal matter (Karapanagiotis et al., 1989).

Low molecular weight compounds such as phenols and alcohols and free moisture are expected to volatilise and make up the liquid product streams, while long chain carboxylic acids, aromatic compounds and alkanes from the polymeric structure of GSS are expected to decompose into short chain fractions, permanent gases and complex aromatic structures (Meng et al., 2017; Oikonomopoulos et al., 2010). The peaks at 1600–1500 and 850-700 cm^{-1} indicate the presence of aromatic and alkene fractions in chars due to aromatisation and dehydrogenation reactions at temperatures above 450 °C pyrolysis temperatures, which corresponds to the results obtained from ^{13}C nuclei magnetic resonance analysis of GSS (Mphahlele, 2020; Paneque et al., 2017). The peaks at 3500–3300, 3350–3010 and 1740–1658 could be attributed to primary amides, primary amines, secondary amines and secondary amides, respectively. The peaks correspond to the fingerprint peak at 1385-1345 cm^{-1} which confirms the presence of nitrogenated compounds. Ultimate analyses indicated the presence of nitrogen in GSS which may form harmful NOx gases or a plant nutrients-rich liquid during pyrolysis. Attributes of organically associated inorganic elements are indicated by multiple-peaks at 990-450 cm^{-1} (Fonts et al., 2009).

The organic structural features of GSS disappeared almost completely due to the release of volatile fractions and functional groups under pyrolysis conditions. Nonetheless, a small peak denoting the development of aromatic structures in the char is observed at 850–700 cm^{-1}. A dominating peak was observed at 1200-800 cm^{-1} caused by Si–O–Si vibrations which are attributed to quartz. Quartz is highly unreactive, as a result it is retained in the chars during pyrolysis. The presence of quartz is observed in all samples and corresponds to the XRF results of their ash samples which are reported elsewhere by Mphahlele (2020). Silicon, which is a main constituent in quartz was reported in large amounts in the ashes of the feed materials. Upon heating, drying of the GSS sample at ~500 °C, a complete expulsion of water is expected. However, a peak at 3700-3600 cm^{-1} representing the water of hydration in the IR spectrum of the char sample, possibly due to the hygroscopic water from the environment during cooling.

3.2. TGA analysis

Fig. 2a and b illustrate the thermal decomposition profiles of GSS at different heating rates (10, 20 & 30 °C/min). The DTG results obtained at 30 °C/min show a clear distinction of peaks as a function of temperature; therefore, this experiment was used to explain the pyrolysis characteristics of GSS. Thermal decomposition of GSS occurred in three stages. The first stage was the dehydration stage which occurred from 30 to 150 °C with a mass loss of 1.83 wt %. The mass loss in the first stage is ascribed to the loss of free moisture and light volatile compounds such as encapsulated aliphatic and aromatic structures. Braga et al. (2014) stated that biomass with moisture less than 10 wt% is suitable for utilisation in the thermochemical process (pyrolysis, combustion or gasification) indicating suitability of GSS in pyrolysis units (Braga et al., 2014).

The second stage was the main devolatilisation stage (49.68 wt%) which occurred at temperatures within the range of 150–570 °C/min indicated the presence of organic fractions such as lipids, cellulose and

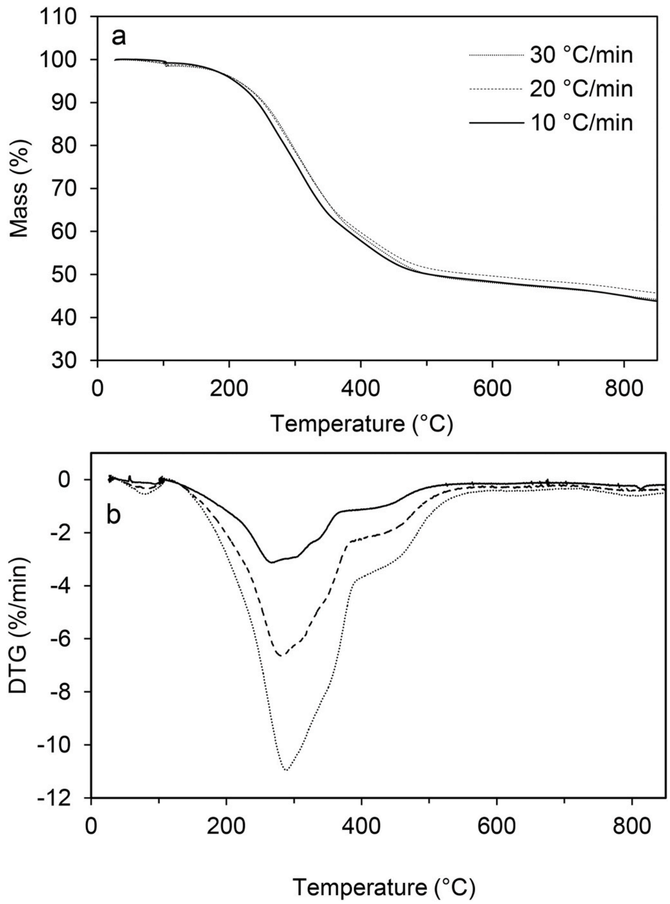

Fig. 2. TGA (a) and DTG (b) thermal profiles obtained from SS TGA analysis at various heating rates.

protein decomposition. Bach and Chen (2017) reported decomposition of lipids and protein to take place at 150–550 °C and 210–310 °C, respectively. Also, the cellulosic materials were found present in the GSS samples decomposing at a temperature between 240 and 400 °C (Lin et al., 2016). The devolatilisation of these organic fractions overlap during pyrolysis which explains why the DTG does not reach zero decomposition rate in between regions which are ascribed to the breakdown of lipids, protein and cellulose in the main devolatilisation stage.

The third stage (stable or slow pyrolysis stage) involved decomposition of carbonaceous residues and decomposition of salts of carboxylic acids from 540 to 750 °C which accounts for 2.43 wt% mass loss (Tsemane et al., 2019). After this stage (750–850 °C), a small peak with mass loss of 1.87 wt% is observed on the DTG graph attributing to interactions of non-mineral inorganics (Folgueras et al., 2013; Mphahlele, 2020).

3.3. Influence of heating rate

From the TGA curves in Fig. 2a it is shown that increasing β from 10 to 30 °C/min slightly effected the decomposition profile of GSS, possibly due to the narrow β intervals which also falls far below the β for fast pyrolysis (β > 100 °C/min). However, the DTG curves (Fig. 2b) and the pyrolysis characteristic parameters in Table 3 show that an increase in β resulted in a shift of the decomposition reactions towards higher temperature regions, especially in the second and third stages. The peak decomposition temperature rose from 266.78 °C to 284.93 °C, and the highest decomposition rate of 1.10%/min.mg was obtained at 30 °C/min. These findings correspond to those conducted on rice straw (Tao et al., 2020), *Typha latifolia* (Ahmad et al., 2017), cellulose and poly

Table 3
Characteristic parameters for pyrolysis of SS obtained at various heating rates.

β (°C/min)	T_i (°C)	T_m (°C)	T_f (°C)	R_p (%/min.mg)	R_v (%/min.mg)
10	150.23	266.78	533.53	0.32	0.06
20	150.23	277.05	555.87	0.67	0.09
30	150.23	284.93	573.51	1.10	0.12

T_i – temperature at which main decomposition started.
T_m – the maximum decomposition peak temperature.
T_f – temperature at which main decomposition terminated.
R_p – maximum rate of decomposition.
R_v – average rate of decomposition.

(methyl methacrylate) (Özsin, 2020) where an increase in β increased the rate of decomposition (R_p). The change in the pyrolysis characteristics parameters with an increase in β can be associated with heat and mass transfer limitations through the bed of the sample (Font et al., 2005; Seebauer et al., 1997; Shahbeig and Nosrati, 2020). This creates temperature gradients between the surface and the centre of the GSS particles causing generation of intermediate volatile at a rate faster than their diffusion rate from the centre to the surface of GSS (Mallick et al., 2018). It is noteworthy that the heat transfer limitations is insignificant to make the kinetic parameters apparent ones especially that the iso-conversional methods does not take heat transfer phenomena into account.

3.4. Kinetics evaluation: iso-conversional method

The second stage of decomposition illustrated the characteristics of the main decomposition/active pyrolysis stage (Fig. 2a and b). For this reason, the kinetic analysis was only focused on the 150–570 °C temperature range. The linear plots shown in Fig. 3a, b and c were obtained from the kinetic model equations for KAS, FWO and Starink methods, respectively. The linear plots were used to calculate the E_a from the slopes with respect to conversion (x). Conversions between 0.1 and 0.9 were considered for this study and the corresponding E_a values are presented in Table 4. The R^2 ranging from 0.9395 to 0.9979 depending on the method used, signifies the confidence in the adopted kinetic methods.

E_a increased with an increase in x due to decomposition of cellulose, lipids and protein at different temperatures and the occurrence of complex endothermic reactions such as boudouard reactions and decarboxylation of salts of carboxylic acids at 500 °C (Heuer et al., 2019; Udayanga et al., 2019). The induced values of E_a varied between 198.61 and 328.04 kJ/mol, 198.61–338.75 kJ/mol, and 197.70–326.85 kJ/mol for Starink, FWO and KAS, respectively. The variations in E_a are ascribed to thermal decomposition of various GSS constituents which occurs at different temperature ranges (Bach and Chen, 2017; Lin et al., 2016). The individual activation energies of cellulose (220.28 kJ/mol), protein (180.08 kJ/mol) (Qiao et al., 2019) and lipids (112.91 kJ/mol) (Bach and Chen, 2017) and their trends as functions of conversion, suggest that the sudden drop in the overall activation energy of the GSS pyrolysis at $x = 0.3$ is probably due to the predominant protein decomposition at ~280 °C. The averaged E_a value from the current study and that of Ma et al. (2019) and Shahbeig and Nosrati (2020) varied significantly which could possibly be due to the distinct nature and composition of the GSS sample taken from different demographically situated wastewater treatment plants.

3.5. Model fitting: Coats-Redfern results

Table 5 presents the kinetic parameters of the main decomposition stage (150–570 °C) obtained by fitting different kinetic models in accordance with the Coats-Redfern technique. E_a varied with the type of model employed. The highest activation energy was obtained from the third-order (R3) reaction model (16.29 kJ/mol) followed by three-

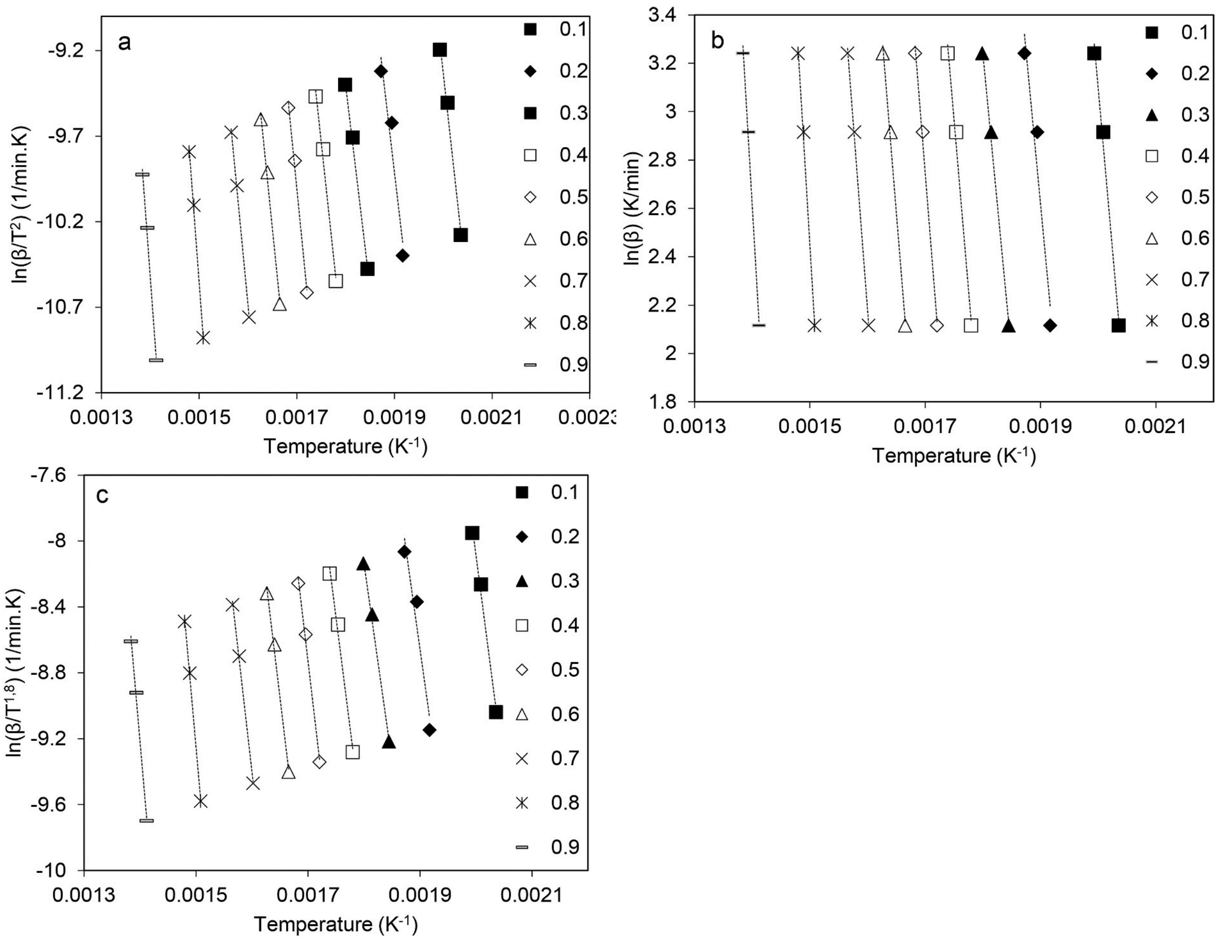

Fig. 3. Linear correlation plots for determining pyrolysis E_a values using a) KAS, b) FWO and c) Starink.

Table 4
Kinetic parameters of SS pyrolysis obtained by three iso-conversional methods.

x	KAS		FWO		Starink	
	Ea (Kj/mol)	R^2	Ea (Kj/mol)	R^2	Ea (Kj/mol)	R^2
0.1	216.13	0.991	224.39	0.9917	216.95	0.9911
0.2	200.45	0.9395	209.23	0.9443	201.33	0.9400
0.3	197.70	0.9957	198.61	0.9961	198.61	0.9958
0.4	221.35	0.9922	230.80	0.9929	222.29	0.9923
0.5	239.09	0.9948	248.85	0.9952	240.06	0.9948
0.6	233.54	0.9944	243.64	0.9949	234.55	0.9944
0.7	248.25	0.9977	258.75	0.9979	249.30	0.9977
0.8	319.41	0.9976	330.54	0.9978	320.52	0.9977
0.9	326.85	0.9944	338.75	0.9948	328.04	0.9945
AVG	218.04		225.92		218.97	

dimensional diffusion (D3) reaction model implying R3 and D3 models played a major role during pyrolysis reactions. R^2 for R3 and D3 was 0.9460 and 0.9760 respectively, which indicated confidence in the values of E_a obtained from these reaction models. Slightly lower values of E_a were obtained with R1, R2, D1, D2, D4 and F5 with acceptable R^2 values ranging from 0.9147 to 0.9879. The low values of E_a along with R^2 slightly above 0.9 illustrate that a single model cannot be used to define the pyrolysis reaction mechanism of complex material such as GSS. Integrated use of these reaction models may be essential to accurately define the solid-state reaction mechanism(s) of the GSS pyrolysis.

The values of E_a obtained from different models were significantly lower than those obtained from the iso-conversional methods. As a result, the Coats-Redfern approach cannot be used to determine kinetic parameters of the GSS pyrolysis confidently. Also, the variations of E_a

Table 5
Single heating rate reaction models with integral and differential functions.

Model/Code	Ea (Kj/mol)	R^2	A
R1	6.81	0.9793	16.14
R2	11.56	0.9879	273.63
R3	17.46	0.9460	6980.62
D1	12.08	0.9147	59.93
D2	13.93	0.9497	87.84
D3	16.29	0.9790	73.35
D4	14.70	0.9616	30.26
P2/3	7.75	0.8769	11.49
P2	−0.90	0.3092	−0.12
P3	−2.34	0.8245	−0.21
P4	−4.25	0.8910	−0.23
A1	2.80	0.9181	1.70
A2	0.79	0.5204	0.24
A3	−1.21	0.7542	−0.19
A4	−2.21	0.9191	−0.24
F4	2.93	0.5225	6.25
F5	11.08	0.7639	1274.81
F1	6.82	0.9798	16.23
F2	4.95	0.8986	2.35
F3	−15.55	0.9469	−0.002

suggest that pyrolysis of GSS involves multiple parallel reactions of different reaction mechanisms taking place simultaneously.

Similar results were reported by Koçer et al. (2020) for the pyrolysis of *Chlorella Vulgaris*. Zhao et al. (2018) and Gao et al. (2013) analysed kinetic models for pyrolysis of sewage sludge and hazelnut blend, and tobacco residues, respectively. Although diffusion and order-based models show good correlation, variations in E_a were reported, which was also the case in the current study. However, Açıkalın (2011) reported a good agreement between the E_a values obtained from Coats-Redfern and iso-conversional methods from the pyrolysis of walnuts.

3.6. Reaction mechanism by master plots

The reaction mechanism of the GSS thermal decomposition was studied by adopting the Criado (1978) method. This method makes use of the DTG data to develop a series of curves for quick and simple assessment of reaction mechanisms. The experimental results were obtained by DTG analysis of GSS at a heating rate of 10 °C/min, and the focus was based on the main decomposition (130–570 °C) zone with conversion degrees from 0.1 to 0.9 because this is where reactions that determine the composition and yields of pyrolysis products. Fig. 4 illustrates the comparison of master plots corresponding to different reaction mechanism their corresponding f(x) and g(x) functions as indicated in Table 6.

Reaction mechanism at $0 \le x \le 0.5$: In this region, there is no reaction mechanism that confidently matches that of the experimental curve. At $x = 0.1$ conversion, the experimental curve and the 1–4 dimensional nuclei growth models by Avrami-Erofeev (A1, A2, A3 and A4) intercepted which could imply decomposition of GSS depending on the growth of the gem nuclei.

However, at $x = 0.2$, a point of intersection between the master plot of the experimental model and the R2 model was observed. As the degree of conversion increased, the experimental curve approached the R3 theoretical model to the point of $x = 0.5$. This supports the observations made from the Coats-Redfern approach that, R3 played a major role in describing the reactions that took place during the main stage of the GSS pyrolysis. Order based models are developed on a simple assumption that the rate of reaction {k(T)} was mainly dependent on conversion or concentration of the reactants (Dhyani and Bhaskar, 2018).

Reaction mechanism at $0.5 \le x \le 0.9$: A dominating relation between the experimental and the z-master plots of R2 was obtained in this region. However, at 0.7, the experimental curve crossed R2 theoretical curve approaching P2/3, P2, P3 and P4 theoretical curves and rapidly ascended to R1, A1, A2, A3, A4 and A5 theoretical curves. This suggests a major change in reaction mechanism as the volatile fraction feedstock was exhausted at a higher conversion. The iso-conversional methods show a spike in E_a at this conversion. The shoulder-like change in the profile of the experimental curve is also reported by (Dhyani et al., 2017; Kumar et al., 2020).

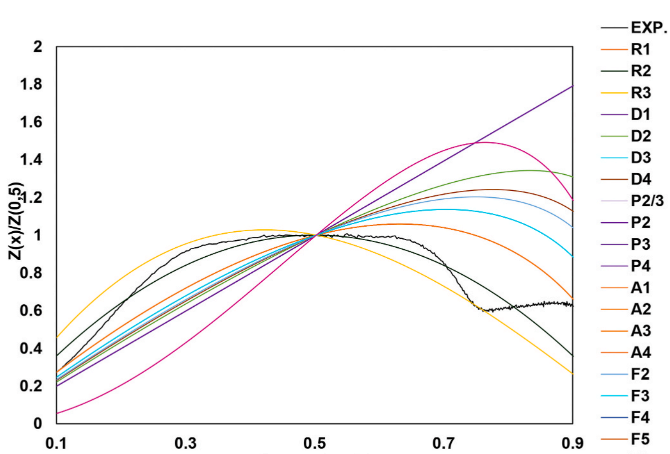

Fig. 4. Experimental and theoretical Z-master plots for correlation of reaction mechanism for the GSS pyrolysis at 30 °C/min.

Table 6
Thermodynamic parameters for SS pyrolysis obtained at 30 °C/min.

x	A (s^{-1})	ΔH (kJ/mol)	ΔG (kJ/mol)	ΔS (J/mol)
KAS				
0.1	6.04×10^{18}	211.49	155.07	101.10
0.2	1.91×10^{17}	195.81	155.42	147.95
0.3	1.04×10^{17}	193.06	155.48	137.63
0.4	1.91×10^{19}	216.71	154.96	226.20
0.5	9.40×10^{20}	234.45	154.60	292.47
0.6	2.78×10^{20}	228.90	154.71	271.75
0.7	7.03×10^{21}	243.61	154.43	326.67
0.8	4.14×10^{28}	314.77	153.26	591.61
0.9	2.11×10^{29}	322.21	153.15	619.26
AVG	2.07×10^{20}	213.40	155.04	196.18
FWO				
0.1	3.71×10^{19}	219.75	154.90	116.20
0.2	1.32×10^{18}	204.59	155.22	180.84
0.3	1.27×10^{17}	193.97	155.46	141.06
0.4	1.52×10^{20}	226.16	154.77	261.50
0.5	8.02×10^{21}	244.21	154.42	328.90
0.6	2.56×10^{21}	239.00	154.51	309.48
0.7	7.05×10^{22}	254.11	154.24	365.83
0.8	4.72×10^{29}	325.90	153.10	632.97
0.9	2.84×10^{30}	334.11	152.99	663.44
AVG	1.80×10^{21}	221.28	154.88	223.00
Starink				
0.1	7.24×10^{18}	212.31	155.05	102.60
0.2	2.32×10^{17}	196.69	155.40	151.25
0.3	1.27×10^{17}	193.97	155.46	141.06
0.4	2.34×10^{19}	217.65	154.94	229.71
0.5	1.16×10^{21}	235.42	154.58	296.10
0.6	3.47×10^{20}	229.91	154.69	275.54
0.7	8.85×10^{21}	244.66	154.41	330.57
0.8	5.29×10^{28}	315.88	153.24	595.75
0.9	2.73×10^{29}	323.40	153.13	623.67
AVG	2.57×10^{20}	214.33	155.02	199.38

3.7. Thermodynamic properties

The values of A, ΔH, ΔG and ΔS were calculated for each conversion for the experiments conducted at 30 °C/min using the corresponding values of activation energy as presented in Table 6. A is a measure of the number of collisions of molecules required for a reaction to occur over time (Shahbeig and Nosrati, 2020). It indicates the reaction chemistry whereby values of A less or equal to 10^9 s^{-1} imply that the reaction is surface dependent. On the other hand, values of A $>10^9$ s^{-1} suggest a simple reaction complex (Yuan et al., 2017). For the present work, the values of A during the GSS pyrolysis ranged from 10^{18} to 10^{30} at 0.1 to 0.9 conversions, indicating the occurrence of complex reactions. The occurrence of complex reactions can be ascribed to the complex chemical structure of GSS corresponding to the multiple decomposition peaks on the DTG curves. These results are in good agreement with those of other studies (Shahbeig and Nosrati, 2020).

Enthalpy change also referred to as enthalpy of formation, is the amount of energy consumed by reactants to form products (Kotz et al., 2008). In this case, it is the amount of energy consumed by GSS to form volatiles and char. The difference between E_a and ΔH is less than 5 kJ/mol suggesting the absence of a potential energy barrier for the formation of the GSS pyrolysis products (Liu et al., 2020). The values of ΔH were positive showing that the reaction was mainly endothermic with averages of 213.40, 221.28 and 214.33 kJ/mol for KAS, FWO and Starink, respectively. ΔH varied from 193.06 to 334.11 kJ/mol, depending on the method used, with conversions indicating the change in reaction mechanisms due to the different thermal decomposition temperatures of the GSS constituents. The amount of energy that is made available for utilisation after the reaction is called the Gibbs free energy. Average values of ΔG, the maximum amount of mechanical energy that can be obtained from a specific mass of material, are 154.08, 155.88 and 154.02 kJ/mol for KAS, FWO and Starink, respectively. A slight change in the ΔG was observed as conversion increased indicating lack of free or surplus energy with respect to the conversion change. The values of ΔG in this study are slightly lower than those of *Typha latifolia* (171–176 kJ/mol) (Ahmad et al., 2017), cattle manure (161.01–166.23 kJ/mol) (Yuan et al., 2017) and sewage sludge (155.96–163.13 kJ/mol) studied by (Shahbeig and Nosrati, 2020). ΔS shows the proximity of a system to its thermodynamic equilibrium, whereby a positive value shows a higher disorder of products than the reactants through bond dissociation (Yuan et al., 2017). High values of ΔS (101.10–619.26 J/mol), (116.20–663.44 J/mol) and 102.60–623.67 J/mol) indicate a high reactivity of the sample. The values of ΔS do not agree with those obtained by Naqvi et al. (2019). Naqvi et al. (2019) reported negative values of ΔS which indicates gentle reactions in the activated state possibly due to a relatively higher composition of mineral matter which are less reactive at temperatures below 600 °C (Naqvi et al., 2019). The variations in the values of the thermodynamic properties of the GSS pyrolysis indicates complex reaction chemistry (Gokul et al., 2019; Shahbeig and Nosrati, 2020; Yuan et al., 2017).

4. Conclusions

In this study, the thermal decomposition characteristics, kinetics and thermodynamic properties of GSS were investigated by conducting TGA experiments during pyrolysis. GSS is a complex material with a wide range of organic functional groups with organically associated inorganic compounds. Fourier transform infrared spectroscopy (FTIR) analysis results indicates disappearance of volatile organic compounds and formation of aromatic structures during the GSS pyrolysis. During pyrolysis, organic compounds volatilised and formed gaseous and liquid pyrolytic products. Pyrolysis of GSS occurred in three stages, namely, dehydration (30–150 °C), main devolatilisation/active pyrolysis (150–550 °C) and the stable zone (550–800 °C). The kinetic model-free methods can be used (KAS, FWO and Starink) to predict pyrolysis E_a of GSS with satisfactory values of R^2 greater than 0.94. Pyrolysis activation energy increased with an increase in conversion, except at $x = 0.3$ due to the lipids decomposition. The thermodynamic properties indicate complex reaction chemistry and that the semi-volatile and volatile organic compounds contained in the GSS sample volatilised at lower and elevated temperatures rapidly during pyrolysis. The use of a modelled kinetic method to describe the reaction mechanism and determine the values of E_a and A was unsuccessful. Pyrolysis of GSS involves complex reaction chemistry with multiple parallel reactions which follow different reaction mechanisms. Third order and three-dimensional (3D) diffusion-reaction mechanisms dominate the reaction mechanisms of the GSS pyrolysis. The use of z-master plot can localise the dominance of reaction models with a degree of conversion. Results of this study can be useful for reaction performance predictions and global energy pyrolysis process simulations and optimisation Future work could involve the use of comprehensive kinetic models to describe the global sewage sludge pyrolysis mechanism.

Author contribution

Katlego Mphahlele: Visualization, Investigation, Validation, Conceptualization, Methodology, Software, Writing – original draft preparation, Ratale Matjie: Data curation, Visualization, Validation Supervision, Writing – original draft preparation. Peter Osifo: Supervision, Writing- Reviewing and Editing,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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